Simulation Study: Base Model and Sensitivity Scenarios

A Bayesian Framework for Causal Inference with Error-Prone Environmental Exposures

Johnson Vo

September 22, 2026

1 Motivation

The simulation reflects the two-level structure of the application: area-level count health outcomes for participants nested within Forward Sortation Areas (FSA or the first three characters of a Canadian postal code), with area-level exposure assigned from a spatiotemporal prediction model.

2 Notation

- $i = 1, \ldots, I$: area index (FSAs)
- A_i: latent true area-level $PM_{2.5}$ exposure, unobserved
- Z_i: observed error-prone surrogate from a spatiotemporal prediction model (e.g. CanOSSEM)
- $\mathbf{X}_i = (X_{i1}, \ldots, X_{i4})^\top$: area-level confounders affecting both exposure and outcome
- D_i: area-level calibration covariate explaining prediction error magnitude in Z_i (e.g. distance to nearest monitoring station, area deprivation)
- Y_i: area-level count health outcome (number of incident depression cases)
- N_i: area population size, used as Poisson offset
- ϕ_i: spatial random effect on the latent exposure surface, following the CAR prior in equation (1)
- $\mathbf{W}$: $I \times I$ binary adjacency matrix ($W_{ij} = 1$ if areas i and j share a border)
- $\mathbf{D}_W = \text{diag}(\mathbf{W1})$: diagonal degree matrix (number of neighbours per area)
- $\rho_A \in [0, 1)$: spatial correlation in the exposure CAR prior
- σ^2_ϕ: marginal variance of the spatial random effect ϕ_i

As in the thesis proposal, $\mathbf{X}_i$ controls confounding while D_i indexes the measurement error mechanism. D_i may overlap with $\mathbf{X}_i$ but is conceptually distinct.

3 Key Assumptions

The simulation satisfies the following identification assumptions from the thesis proposal.

- **Non-differential surrogate:** $Y_i \perp Z_i \mid A_i, \mathbf{X}_i, D_i$. Once we condition on the true exposure and covariates, the surrogate carries no additional information about the outcome. Satisfied by construction since Y_i is generated from A_i only and Z_i never appears in the outcome equation.
- **Transportability:** The error relationship is assumed transportable to the target population. When direct calibration data are unavailable, inference about A_i is more model-dependent; this is examined in sensitivity scenario S2.

4 Base Model

4.1 Spatial Structure and CAR Prior

Areas are arranged on a $\sqrt{I} \times \sqrt{I}$ synthetic grid with rook contiguity (shared edges only). For this simulation chosen values of $I \in \{196, 400\}$ were utilized.

Following thesis proposal Section 4.3, spatial dependence enters only through the latent exposure model via ϕ_i. The spatial random effect follows the Conditional Autoregressive (CAR) prior:

$$\boldsymbol{\phi} \mid \rho_A, \sigma_\phi^2 \sim \mathcal{N}\big(\mathbf{0},\ \sigma_\phi^2(\mathbf{D}_W - \rho_A \mathbf{W})^{-1}\big)\,. \tag{1}$$

- ϕ_i is a spatially structured random effect: neighbouring areas have similar ϕ_i values, inducing correlation in their true exposures A_i
- The precision matrix $(\mathbf{D}_W - \rho_A \mathbf{W})$ is sparse with adjacent pairs having off-diagonal entry $-\rho_A$ and non-adjacent pairs having zero
- $\rho_A = 0$: areas are spatially independent; $\rho_A \to 1$: strong smoothing across neighbours
- Spatial correlation in the outcome is induced indirectly through A_i; no separate outcome random effect is added

4.2 Covariates and Population Sizes

- **Area confounders:** $X_{ik} \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$ for $k = 1, 2, 3, 4$
- **Population sizes:** $\log(N_i) \sim \mathcal{N}(8.5, 1.2^2)$, giving median $\approx$ 4,900, 5th percentile $\approx$ 600, 95th percentile $\approx$ 40,000. Motivated by Statistics Canada 2021 Census: Ontario has $\approx$ 500 FSAs with total population 14,223,942, ranging from dense urban FSAs in Toronto to sparse rural FSAs in Northern Ontario.

4.3 Calibration Covariate D_i

D_i captures area characteristics that determine prediction error in Z_i.

Base model: scalar D_i. We generate a raw covariate $D_i^* \overset{\text{iid}}{\sim} \text{Gamma}(2, 1)$ (mean 2, variance 2) representing distance to the nearest monitoring station, then standardise:

$$D_i = \frac{D_i^* - 2}{\sqrt{2}},$$

so that D_i has mean 0 and variance 1. Standardisation prevents numerical blow-up in the exponential error variance. The interpretation of η_1 demonstrates that a one standard deviation increase in monitor distance multiplies the error variance by $\exp(\eta_1)$.

- Distance is strictly positive and right-skewed and hence a Gamma distribution naturally showcases this. Most Ontario FSAs are moderately close to one of the 34 monitoring stations, but some rural areas are very far.
- **Motivation:** Gryparis et al. (2009) and Szpiro et al. (2011) show that prediction error variance is primarily determined by distance to the monitor network. MELONS (Evangelopoulos et al., 2026) identified distance, deprivation, and housing type as empirical determinants of $PM_{2.5}$ measurement error in London, UK

Extension: vector D_i. In a full implementation D_i would be a vector with components such as: $D_{i1} \sim \text{Gamma}(2, 1)$ (distance to monitor, standardised), $D_{i2} \sim \mathcal{N}(0, 1)$ (income and employment indicators as a standardised score), $D_{i3} \sim \text{Beta}(2, 2)$ (proportion detached housing). The error model would be $\omega^2(\mathbf{D}_i) = \exp(\eta_0 + \eta_1 D_{i1} + \eta_2 D_{i3})$. Left as a future extension.

4.4 Latent True Exposure

The latent true exposure is generated from:

$$A_i \mid \mathbf{X}_i, \phi_i \sim \mathcal{N}(7 + 1.5X_{i1} - 1.5X_{i2} - 1.5X_{i3} + 1.5X_{i4} + \phi_i,\ 9)\,, \tag{2}$$

with $\sigma_A = 3$, where ϕ_i is the spatial random effect following the CAR prior defined in equation (1) with parameters (ρ_A, σ_ϕ^2). The spatial random effect ϕ_i induces correlation in A_i across neighbouring areas: areas with ρ_A close to 1 have exposures that vary smoothly across the grid, reflecting a regional wildfire plume, while $\rho_A = 0$ gives spatially independent exposures.

- **Mean = 7 μg/m^3:** Empirical annual average $PM_{2.5}$ in Ontario (mean $\approx$ 7, median $\approx$ 5.4 μg/m^3)
- $\sigma_A = 3$**:** Reflects realistic spatial variation
- **Covariate coefficients** ± 1.5**:** Provides meaningful confounding given the exposure distribution
- **Note on skewness:** Ontario $PM_{2.5}$ is heavily right-skewed (skewness $\approx$ 9.2, 99th percentile $\approx$ 26.4, extreme values $>$ 200 μg/m^3 during severe wildfire events). The Normal captures typical days but not extreme events.

Choosing σ^2_ϕ via R^2 decomposition. With the exposure model now defined, we can determine σ^2_ϕ by targeting a meaningful proportion of the total variance in A_i. The total variance decomposes into three sources: the covariate contribution $\text{Var}(\mathbf{X}_i \text{ terms}) = 4 \times 1.5^2 = 9$, the spatial contribution $\text{Var}(\phi_i) \approx \sigma^2_\phi / \bar{d}$ where $\bar{d} \approx 3.6$ is the average number of neighbours on the grid, and the residual $\sigma^2_A = 9$. The proportion of variance explained by the spatial random effect is:

$$R^2_\phi = \frac{\sigma^2_\phi / \bar{d}}{\text{Var}(A_i)} = \frac{\sigma^2_\phi / \bar{d}}{9 + \sigma^2_\phi / \bar{d} + 9}.$$

Setting a target R^2_ϕ and solving gives $\sigma^2_\phi = \bar{d} \times R^2_\phi \times 18/(1 - R^2_\phi)$. At the previously used value of $\sigma^2_\phi = 0.3$, the implied $R^2_\phi \approx 0.5\%$ — essentially no spatial contribution. We therefore parameterise through the target $R^2_\phi \in \{0.10, 0.30, 0.50\}$ and compute the implied σ^2_ϕ accordingly.

4.5 Measurement Error Model

Two parameterisations of the measurement error model are considered. Both generate Z_i as a function of the true exposure A_i and calibration covariate D_i, and both satisfy the non-differential assumption by construction since Z_i does not appear in the outcome equation.

In both versions the measurement errors $\varepsilon_i = Z_i - \mathbb{E}[Z_i \mid A_i, D_i]$ are assumed conditionally independent across areas given (A_i, D_i). This is a simplifying assumption: Gryparis et al. (2009) show that prediction errors from spatial models are spatially correlated because nearby areas share the same monitoring station data when their surrogates are generated. Incorporating spatially correlated errors is listed as a potential extension.

Version A: Classical error with Di-dependent mean (misspecification scenario). Following Wu et al. (2019), D_i enters through the mean of the surrogate, introducing systematic directional bias:

$$Z_i \mid A_i, D_i \sim \mathcal{N}(\beta_{Z1} A_i + \beta_{Z2} D_i,\ \sigma^2_Z). \tag{3}$$

This model is not fully identifiable when A_i is latent: β_{Z1} and σ^2_Z cannot be separately recovered from the marginal distribution of Z_i without validation data.

- β_{Z1}: attenuation. $\beta_{Z1} = 1$ centres the surrogate on the truth; $\beta_{Z1} < 1$ attenuates high exposures. Wu et al. (2019) use $\beta_1 = 0.8$ as their default.
- β_{Z2}: directional bias from D_i. When $\beta_{Z2} > 0$ the surrogate overestimates true exposure in high-D_i areas (far from monitors), consistent with the MELONS finding of systematic overestimation in deprived and remote areas. When $\beta_{Z2} = 0$ this reduces to standard classical error.
- σ^2_Z: fixed homoscedastic noise. Heterogeneity across areas is captured entirely through the mean shift $\beta_{Z2} D_i$.

Wu et al. (2019) write this in the regression calibration direction as $E(X \mid W, D) = \gamma_0 + \gamma_1 W + \gamma_2^\top D$.

Version B: Classical error with Di-dependent variance (primary model). D_i enters through the variance of the surrogate, relaxing the homoscedasticity assumption of Josey et al. (2023) Assumption 2:

$$Z_i \mid A_i, D_i \sim \mathcal{N}(A_i,\ \sigma_Z^2(D_i)), \qquad \sigma_Z^2(D_i) = \exp(\eta_0 + \eta_1 D_i). \tag{4}$$

- The surrogate is unbiased on average ($E[Z_i \mid A_i] = A_i$) but noisier in high-D_i areas
- $\eta_1 = 0$ recovers Josey's homoscedastic Assumption 2 exactly
- $\exp(\eta_0 + \eta_1 D_i)$ ensures positivity and implies a log-linear (proportional) relationship: a unit increase in D_i multiplies the error variance by $\exp(\eta_1)$, natural for distance-based quantities (Gryparis et al., 2009)
- Unlike Version A, there is no attenuation or directional bias; the error is purely a noise term whose magnitude depends on D_i
- This version is identifiable: the marginal variance of Z_i given $(\mathbf{X}_i, D_i)$ is $\sigma_A^2 + \sigma_\phi^2/\bar{d} + \exp(\eta_0 + \eta_1 D_i)$, allowing estimation of (η_0, η_1) without observing A_i

4.6 Outcome Model

The area-level count outcome follows a Poisson model with a log-linear predictor and population offset, matching Josey et al. (2023):

$$Y_i \mid A_i, \mathbf{X}_i, N_i \sim \text{Poisson}(N_i \exp(\eta_i)), \tag{5}$$

$$\eta_i = -3 + f(A_i) - 0.5X_{i1} - 0.25X_{i2} + 0.25X_{i3} + 0.5X_{i4}. \tag{6}$$

- **Population offset** $\log(N_i)$**:** N_i scales the expected count so that $\exp(\eta_i)$ is the event rate per person. A larger area will have more events simply because it has more people; the offset removes this confound. In R this enters as an additive term $\log(N_i)$ in the linear predictor or via the `offset` argument.
- **Intercept** -3**:** $\exp(-3) \approx 5\%$ baseline event rate, representing baseline depression incidence in the CLSA population (Raina et al., 2019)
- **Confounder coefficients** $\pm(0.5, 0.25)$**:** Follow Josey et al. (2023)
- **ERF** $f(A_i)$**:** Exposure-response function specified in sensitivity scenario S4; three functional forms are evaluated
- **No spatial random effect:** Consistent with thesis proposal Section 4.3; spatial correlation in outcomes is induced indirectly through the spatially correlated A_i surface

Z_i does not appear in equation (5), satisfying the non-differential assumption by construction.

4.7 Causal Estimand

Following Josey et al. (2023), the target estimand is the marginal exposure-response function (ERF):

$$\theta(a) \equiv \mathbb{E}[Y_i(a)],$$

where $Y_i(a)$ is the potential outcome that would occur in area i if, possibly contrary to what is observed, it had received exposure level $A_i = a$. The causal contrast between two exposure levels is $\Delta(a_1, a_0) = \theta(a_1) - \theta(a_0)$. Under the identifying assumptions in Section 3, the ERF is identified by the g-formula:

$$\theta(a) = \mathbb{E}[\mathbb{E}[Y_i \mid A_i = a, \mathbf{X}_i]] \,.$$

- The true $\theta(a)$ is computed by Monte Carlo using each replicate's own simulated covariates $\mathbf{X}_i$ (fresh per replicate, following Josey et al. 2023 `predict_example`), rather than a separate fixed external population. This lets the benchmark truth reflect the sampling variability of the covariate distribution itself alongside the estimators being evaluated
- **Evaluation grid:** 201 equally spaced values covering the bulk of the A_i distribution
- **Primary causal contrast: extreme vs typical exposure:** $\Delta(Q_{99}, Q_{50})$ where Q_{99} and Q_{50} are the 99th and 50th percentiles of the simulated A_i distribution. For the base Normal$(7, 9)$ model these correspond approximately to $Q_{50} \approx 7$ and $Q_{99} \approx 14$, representing the increase in expected depression case count on a severe wildfire day relative to a typical day. This mirrors the empirical contrast between Ontario median $PM_{2.5}$ (≈ 5.4 $\mu g/m^3$) and 99th percentile (≈ 26.4 $\mu g/m^3$). The percentiles are computed from the simulated A_i values in each replicate so that the contrast automatically adapts to the exposure distribution under each scenario.
- **Secondary contrast: IQR:** $\Delta(Q_{75}, Q_{25})$ following standard practice in air pollution epidemiology

5 Simulation Parameter Grid

Table 1: Simulation parameters. Version A and Version B refer to the two error model parameterisations in Section 4. R^2_ϕ denotes the proportion of $\text{Var}(A_i)$ explained by the spatial random effect.

Parameter	Notation	Base / Range	Justification
Design			
Number of areas	I	196	Nearest perfect square to 200
Replicates	R	500	
ERF evaluation points		201	Josey et al. (2023)
Covariates			
Area confounders	X_{ik}	$\mathcal{N}(0, 1)$	Josey et al. (2023)
Population size	$\log(N_i)$	$\mathcal{N}(8.5, 1.44)$	Statistics Canada 2021 Census
Calibration covariate	D_i	Gamma(2, 1), standardised	Monitor distance proxy; Gryparis et al. (2009)
True exposure			
Mean	$\mu_{A,i}$	$7 + 1.5\mathbf{X}_i$	Ontario $PM_{2.5}$ mean ≈ 7 μg/m^3
SD	σ_A	3	Realistic spatial variation
Spatial R^2	R^2_ϕ	$\{0.10, 0.30, 0.50\}$	Proportion of $\text{Var}(A_i)$ explained by spatial effect; 0.30 = base
Spatial correlation	ρ_A	$\{0, 0.5, 0.9\}$	
Error model Version A			
Attenuation	β_{Z1}	0.8	Wu et al. (2019) default
Directional bias	β_{Z2}	$\{0, 1, 2\}$	0 = no bias; > 0 = overestimation
Noise variance	σ^2_Z	1	Fixed moderate noise
Error model Version B			
Baseline variance	η_0	0	$\exp(0) = 1$; moderate baseline error
Di-slope	η_1	$\{0, 0.5, 1.0, 1.5\}$	
Outcome			
Intercept		-3	$\approx 5\%$ baseline event rate (CLSA)
ERF	$f(A_i)$	Linear/Quadratic/Nonlinear	
Confounder coefs	$\boldsymbol{\gamma}$	$(-0.5, -0.25, 0.25, 0.5)$	
Sensitivity scenarios			
D_i^* distribution		Exp(1); empirical	Monotone decreasing; real Ontario data
N_i distribution		Uniform(10, 1000)	Josey comparability
Exposure distribution		Log-normal	Extreme wildfire events
Number of areas	I	$\{100, 196, 400\}$	Scaling; all perfect squares

6 Sensitivity Scenarios

D_i **distribution.** **Gamma(2, 0.5)**: mean 4, more heterogeneity between areas. **Exponential(1)**: monotone decreasing, consistent with Southern Ontario station concentration. **Empirical**: actual FSA-to-NAPS distances from Statistics Canada shapefiles.

Error model alternatives. Version A vs Version B: Data generated under Version A (directional bias, $\beta_{Z2} \in \{0, 1, 2\}$) but analysed with a Version B error model. Quantifies performance degradation when the true error has directional bias that the proposed method does not model. $\beta_{Z2} = 0$ provides an internal consistency check.

Spatial correlation. Scenario B ($\rho_A = 0$): spatially independent exposures. Directly comparable to Josey et al. (2023) which sets $\phi_i = 0$ for all areas. **Scenario A ($\rho_A = 0.9$)**: high spatial smoothing, reflecting a sustained regional wildfire event where nearby FSAs share similar smoke concentrations.

Exposure-response functional forms. Three forms spanning simple to complex, crossed with spatial scenarios to give six main simulation cells.

Linear: $f(A_i) = 0.10(A_i - 7)$. Monotone increasing with no curvature.

Quadratic: $f(A_i) = 0.35(A_i - 7) - 0.035(A_i - 7)^2$. Concave shape that rises then declines within the evaluation window (vertex at $A_i = 12$).

Nonlinear (Josey cosine form): $f(A_i) = 0.25(Z_i + 2) - 0.75\cos(\pi(Z_i + 4)/4) - 0.25 Z_i X_{i1}$, where $Z_i = (A_i - 7) \times 4/5$ if $A_i < 7$ and $(A_i - 7) \times 4/7$ otherwise. Reuses Josey et al. (2023)'s own coefficients directly, rescaled onto our window so the curve stays bounded instead of exploding on the longer right tail. S-shaped curve with an exposure-covariate interaction. The interaction means the ERF shape differs across areas depending on X_{i1}, which BART can capture but parametric GLMs cannot.

Log-normal exposure. $\log(A_i) \sim \mathcal{N}(1.68, 0.54)$ matching Ontario moments: median 5.4, mean 7, SD 5.9. Captures extreme wildfire events (99th percentile ≈ 26 μg/m^3) that the Normal cannot represent.

7 Performance Metrics

For each scenario and method (naive, oracle, proposed, regression calibration):

- **Bias**: $\hat{\theta}(a) - \theta(a)$ averaged over evaluation points and replicates
- **RMSE**: $\sqrt{R^{-1} \sum_r (\hat{\theta}_r(a) - \theta(a))^2}$
- **Coverage**: fraction of 95% credible intervals containing $\theta(a)$
- **Interval width**: mean width of 95% credible intervals
- **Causal contrast bias**: $\hat{\Delta}(Q_{99}, Q_{50}) - \Delta(Q_{99}, Q_{50})$ and $\hat{\Delta}(Q_{75}, Q_{25}) - \Delta(Q_{75}, Q_{25})$, where quantiles are computed from the simulated A_i distribution in each replicate

8 Potential Extensions

- **Spatially correlated measurement errors:** The current model assumes conditional independence of measurement errors $\varepsilon_i = Z_i - \mathbb{E}[Z_i \mid A_i, D_i]$ across areas which is a simplifying assumption that Gryparis et al. (2009) show is violated in practice: prediction errors from spatial models are correlated across nearby areas because those predictions are generated using the same monitoring station data. A natural extension is to give the error term its own spatial covariance structure, either by making D_i spatially structured via a second CAR prior (so that nearby areas have correlated error variances), or by adding an explicit spatially correlated residual $\psi_i \sim \text{CAR}(\rho_\varepsilon, \sigma_\varepsilon^2)$ to the error model.

- **Individual-level binary outcome:** Replacing the area-level Poisson count with individual-level Bernoulli outcomes $Y_{ij} \sim \text{Bernoulli}(p_{ij})$ and individual covariates L_{ij}, matching the CLSA data structure. The Bayesian bootstrap would draw Dirichlet weights over all $N = \sum_i n_i$ individuals rather than over areas.

- **Spatial random effect on outcome (ν_i)**: A second CAR prior capturing residual spatial clustering in health outcomes (Englert et al., 2025).

- **Negative binomial outcome:** The Poisson model assumes $\text{Var}(Y_i) = \mathbb{E}[Y_i]$, which is often violated in practice when area-level counts are overdispersed due to unmeasured heterogeneity across FSAs. A negative binomial outcome relaxes this by introducing a dispersion parameter $\xi > 0$:

$$Y_i \mid A_i, \mathbf{X}_i, N_i \sim \text{NegBin}(\mu_i, \xi), \qquad \mu_i = N_i \exp(\eta_i),$$

  As $\xi \to \infty$ this recovers the Poisson; small ξ gives strong overdispersion. The dispersion parameter $\xi = 1$ is a natural starting point and is the value used by Englert et al. (2025) in their simulation. This extension tests whether the proposed method remains well-calibrated when the outcome model is more variable than the base Poisson.

- **Zero-inflated outcome**: For rare outcomes or small populations where structural zeros arise.

- **Log-normal exposure model:** The base model specifies A_i on the original scale as a Normal with a floor at zero. An alternative motivated by the right-skewed nature of $PM_{2.5}$ concentrations and by the suggestion that $\log(A_i)$ should be regressed on covariates is to model the log-exposure directly:

$$\log(A_i) \mid \mathbf{X}_i, \phi_i \sim \mathcal{N}\big(\gamma_0 + \gamma_1 X_{i1} + \gamma_2 X_{i2} + \gamma_3 X_{i3} + \gamma_4 X_{i4} + \phi_i,\ \sigma^2_{\log A}\big)\,,$$

  so that $A_i = \exp(\log A_i) > 0$ always. The parameters are matched to Ontario $PM_{2.5}$ empirical moments: setting $\gamma_0 = 1.68$ and $\sigma_{\log A} = 0.73$ gives $\mathbb{E}[A_i] \approx 7\ \mu\text{g/m}^3$ and $\text{SD}(A_i) \approx 5.9\ \mu\text{g/m}^3$, matching the provincial monitoring data. The error model in Version B would then operate on the log scale as $\log(Z_i) \mid \log(A_i), D_i \sim \mathcal{N}(\log(A_i), \exp(\eta_0 + \eta_1 D_i))$, giving multiplicative error on the original scale.